

Machine Learning

(RCP23APC301)

Third Year

Semester-VI

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Department of AIML

R C Patel Institute of Technology

(An Autonomous Institute)

Vision

Fostering technical excellence in Artificial Intelligence and Machine Learning through ethics, innovative research, and sustainable education.

Mission

- M1: Innovative and interactive learning processes with high-quality instructional programs in Artificial Intelligence and Machine Learning.
- M2: Fostering a collaborative scientific temper among students with ethical responsibility towards society in AI and ML applications.
- M3: Preparing students from diverse backgrounds to develop aptitude for employment, entrepreneurship, and research in AI and ML with a spirit of professionalism.

About Course:

[illegible]

Prerequisite: Knowledge of *basic probability and statistics, Data Mining and Analytics concepts*

Course Objectives:

- To understand key machine learning concepts: hypothesis formation, bias-variance trade-off, and model evaluation metrics.
- To master regression, classification, and clustering techniques.
- To apply machine learning algorithms to real-world datasets effectively.

Course Outcomes:

Course Outcomes: On completion of the course, the learner will be able to:

CO	Course Outcomes	Blooms Level	Blooms Description
CO1	Explain the types of machine learning and issues in machine learning.	L2	Understand
CO2	Analyse model performance using evaluation metrics.	L4	Analyze
CO3	Implement and tune regression and classification algorithms.	L3	Apply
CO4	Apply knowledge of Bayesian learning principles.	L3	Apply
CO5	Apply association rule mining and clustering techniques.	L3	Apply
CO6	Explain the fundamentals of Neural Networks.	L2	Understand

Unit-I:

Introduction to Machine Learning:

6 Hrs

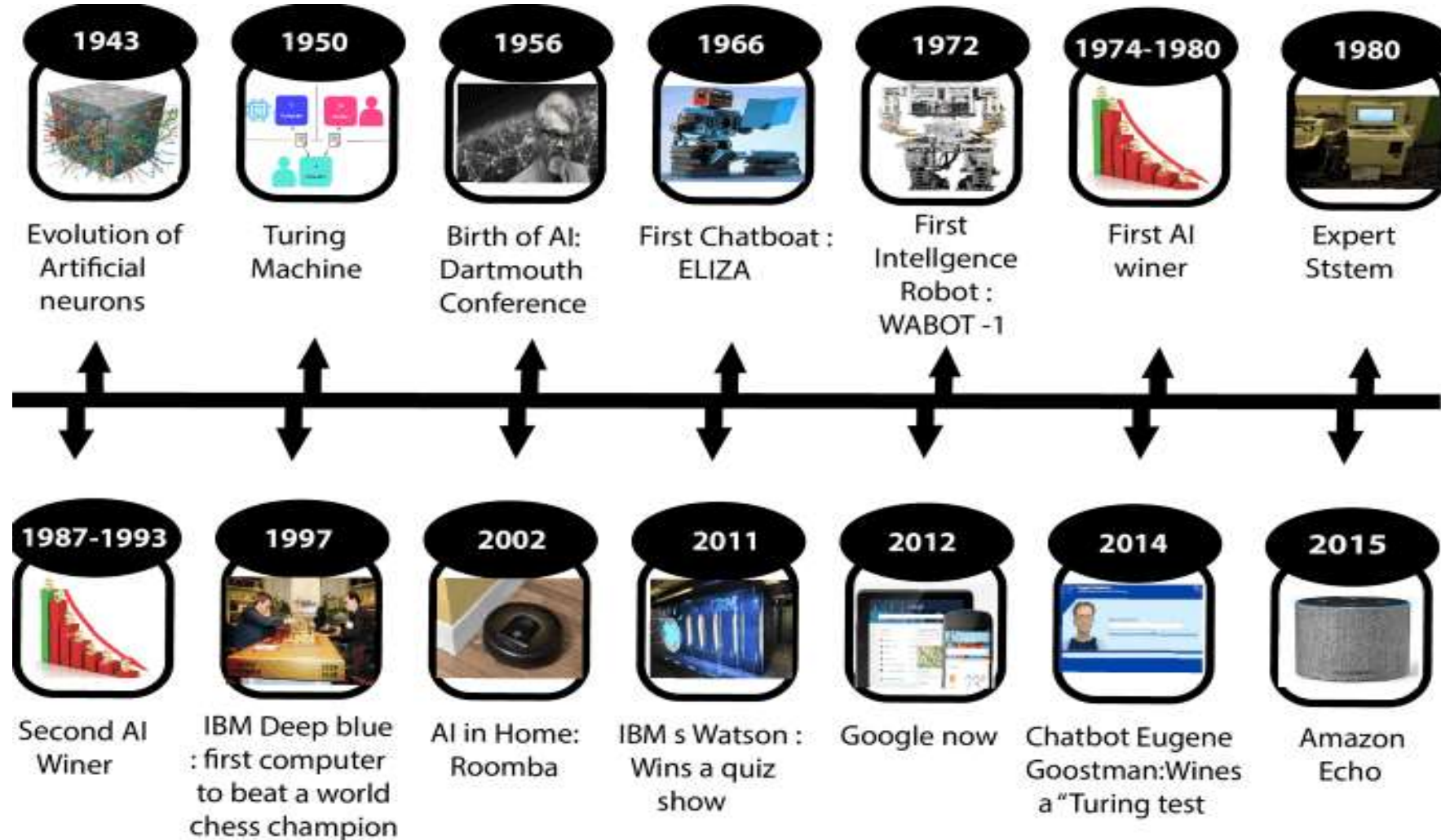
Introduction to Machine Learning

- Types of Machine Learning
- Issues in Machine Learning
- Applications of Machine Learning
- Steps Involved in Developing a Machine Learning Application
- Hypothesis and Inductive Bias
- Bias–Variance Trade-off
- Performance Measures
- Data Validation

Evaluation & Selection

- Metrics for Evaluating Classifier Performance
- Holdout Method
- Random Subsampling
- Cross Validation
- Bootstrap
- Model Selection Using Statistical Tests of Significance
- Comparing Classifiers Based on Cost–Benefit
- ROC (Receiver Operating Characteristic) Curves

Evolution of Artificial Intelligence(AI)



What is Artificial Intelligence?

- Artificial intelligence is an academic discipline was founded in 50s. Actually the “AI” term was coined by [John McCarthy](#), an American computer scientist, in 1956 at The Dartmouth Conference.
- According to John McCarthy, AI is “*The science and engineering of making intelligent machines, especially intelligent computer programs*”.
- Artificial intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think like humans and mimic their actions.

Machine Learning (ML)

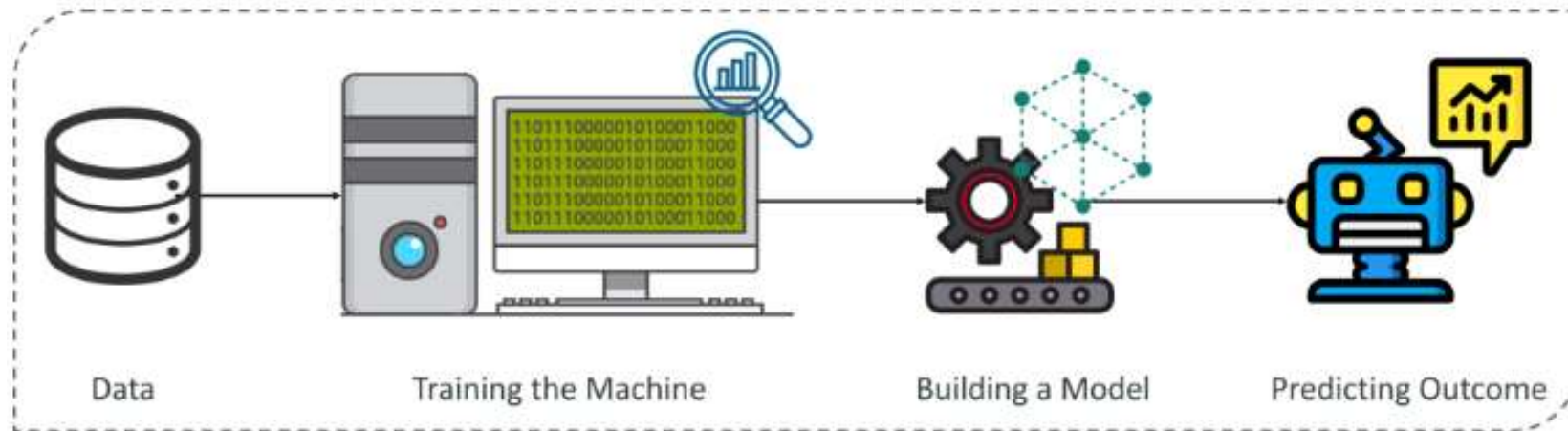
- Machine learning is an application of artificial intelligence that involves algorithms and data that automatically analyze and make decision by itself without human intervention.
- It describes how computer perform tasks on their own by previous experiences.
- Therefore we can say in machine language artificial intelligence is generated on the basis of experience.
- Machine learning - a field of science and a powerful technology that allows machines to learn from data and self-improve.

Machine Learning (ML)

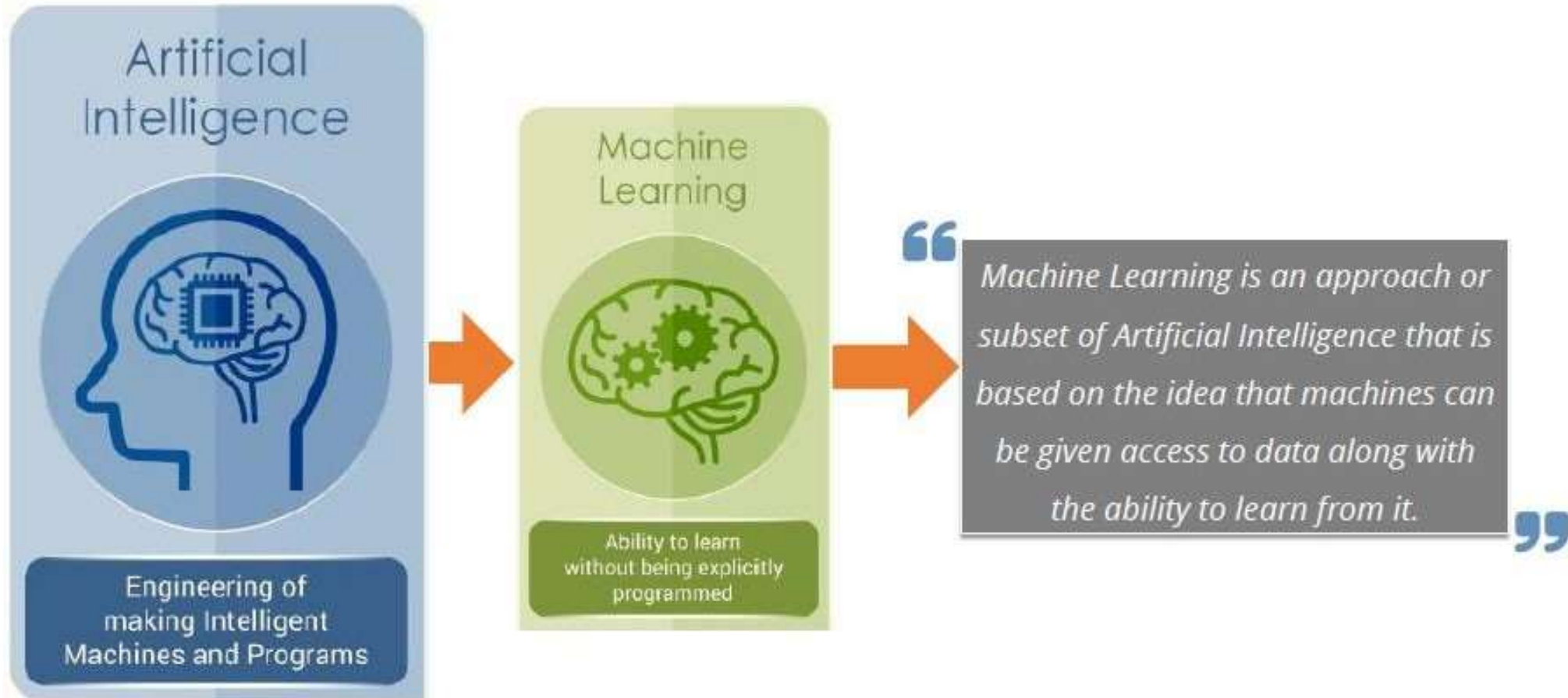
- **Machine learning** is an application of artificial intelligence (AI) that provides systems the ability to automatically learn and improve from experience without being explicitly programmed.
- **Machine learning focuses on the development of computer programs** that can access data and use it to learn for themselves
- **Machine Learning** algorithms automatically build a mathematical model using sample data – also known as “training data” – to make decisions without being specifically programmed to make those decisions.

How does Machine Learning work

- A Machine Learning system **learns from historical data, builds the prediction models, and whenever it receives new data, predicts the output for it.**
- The accuracy of predicted output depends upon the amount of data, as the huge amount of data helps to build a better model which predicts the output more accurately.



Relationship between Artificial Intelligence and Machine Learning



Features of Machine Learning



01

It uses the data to *detect patterns* in a dataset and *adjust program actions accordingly*

It *focuses on the development of computer programs* that can teach themselves to *grow and change* when *exposed to new data*

02



03

It enables computers to *find hidden insights* using *iterative algorithms* without being explicitly programmed

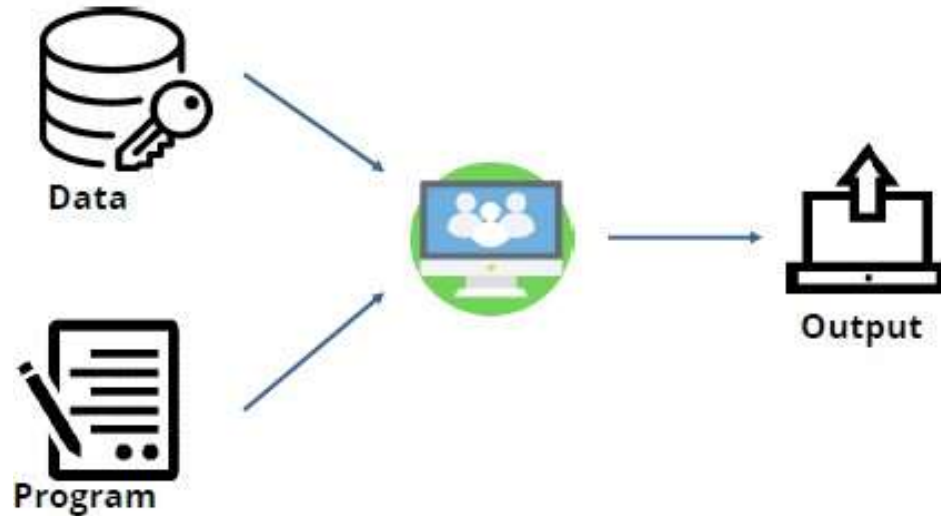
It *automates analytical model building*

04

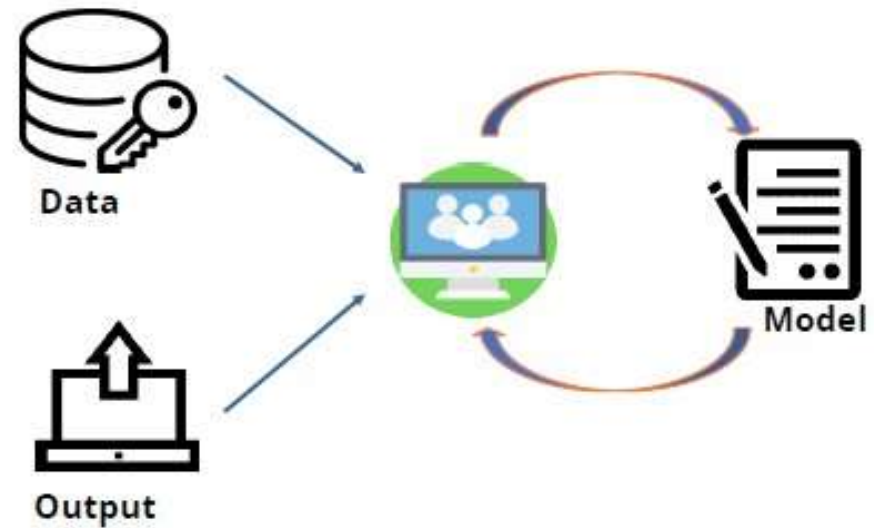


Traditional Approach vs. Machine Learning Approach

Traditional Programming: Data and program is run on the computer to produce the output

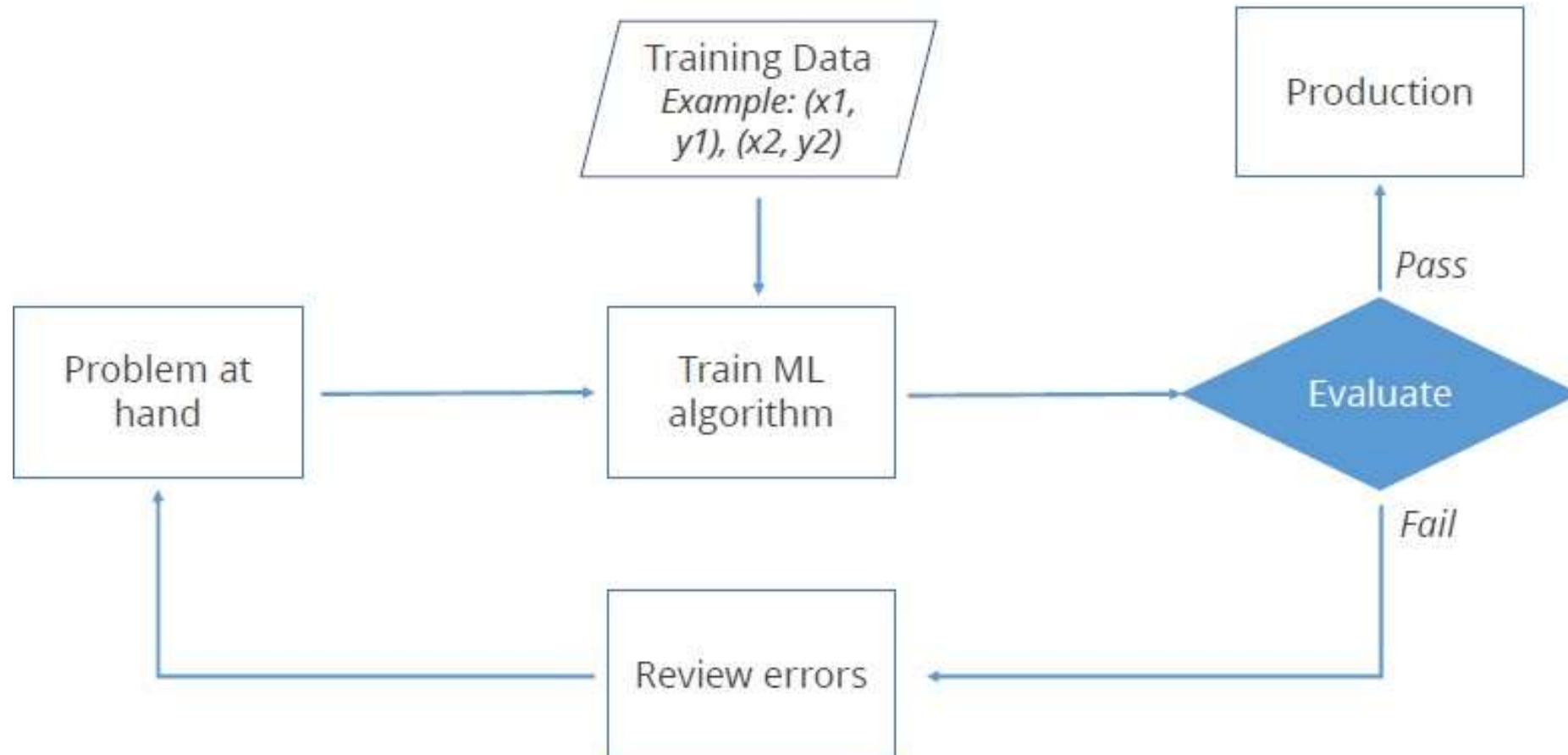


Machine Learning: Data and output is run on the computer to create a program



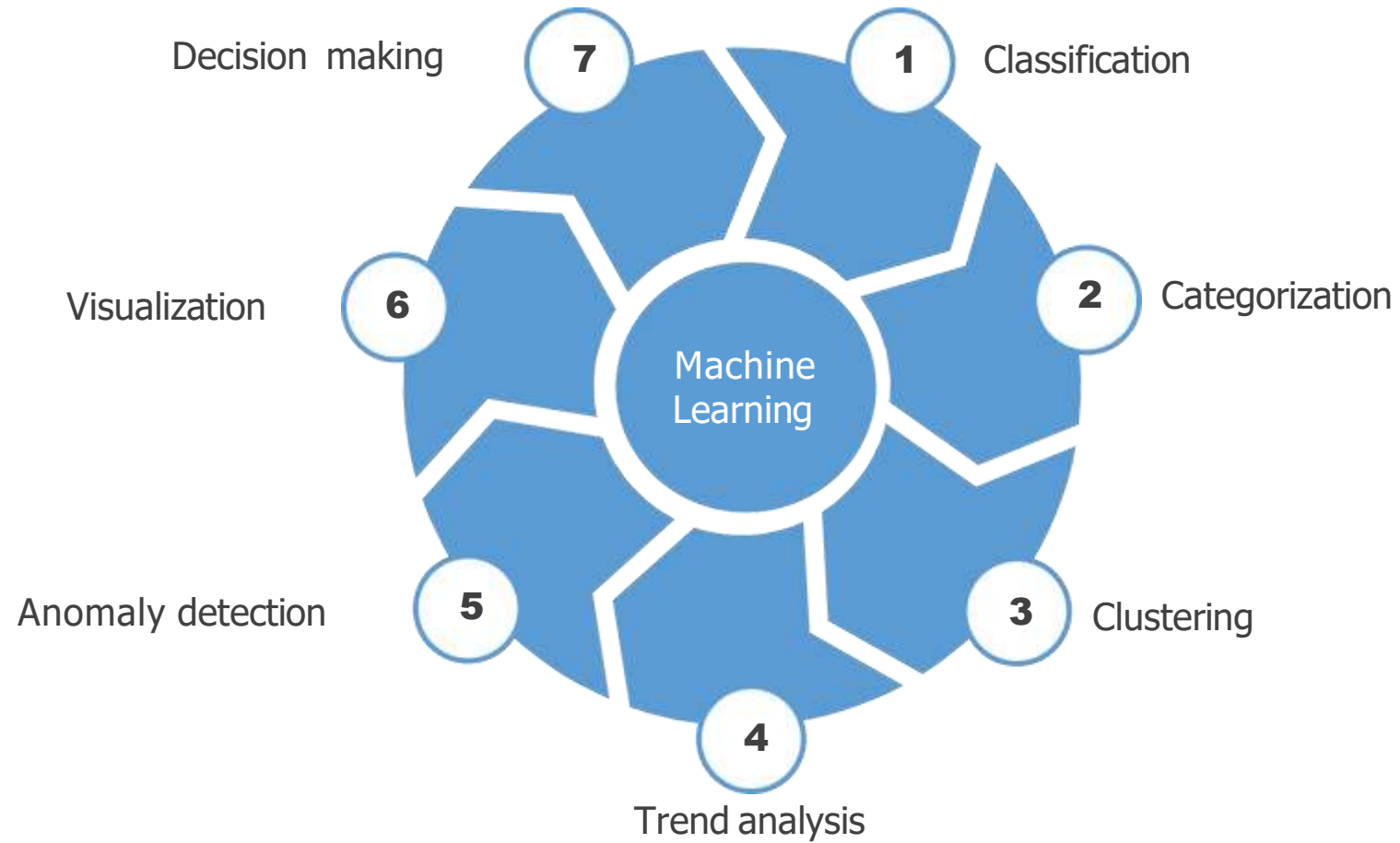
Machine Learning Approach

Machine Learning relies on learning patterns based on sample data.

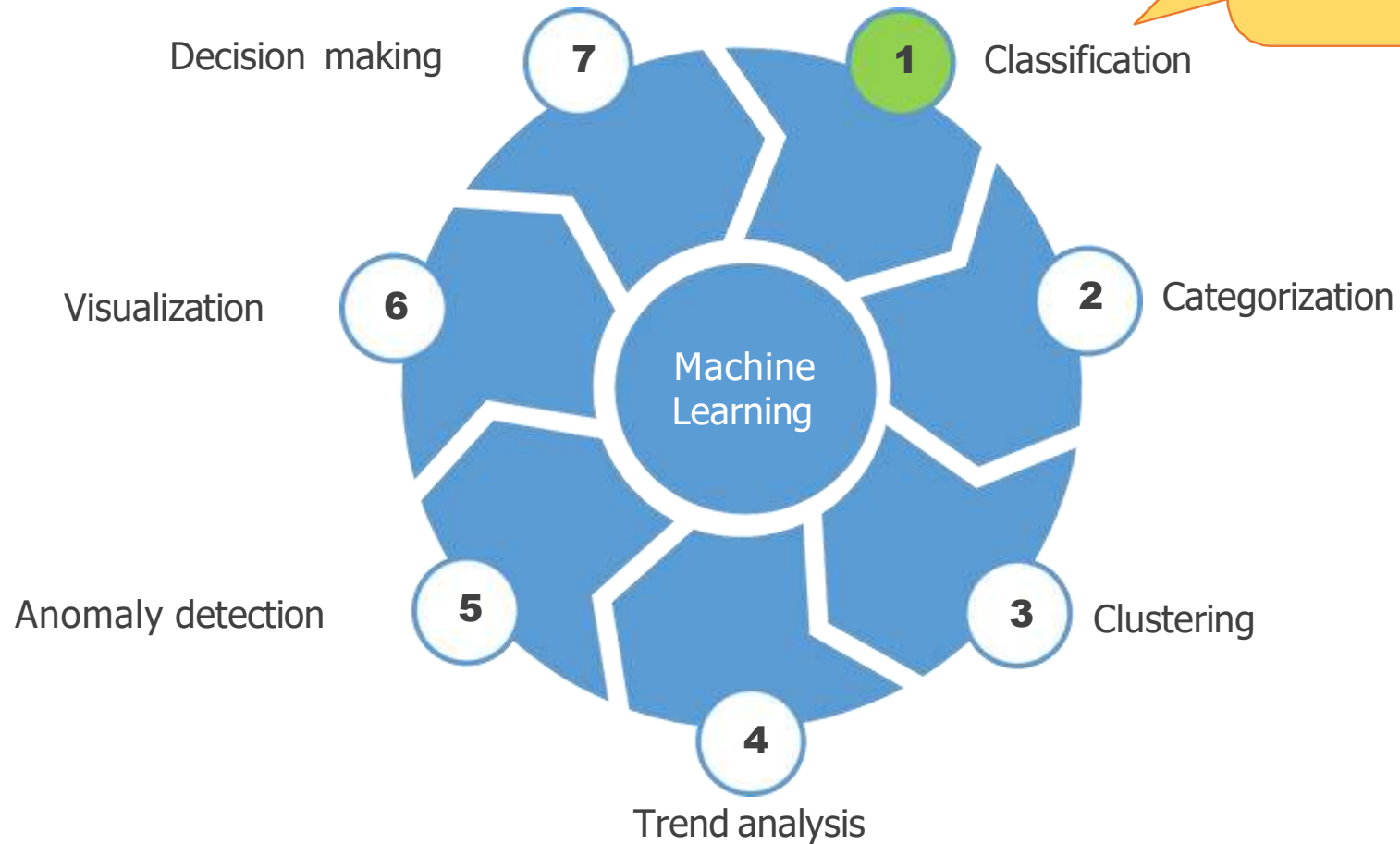


Machine Learning Techniques

Machine Learning uses a number of theories and techniques from Data Science:

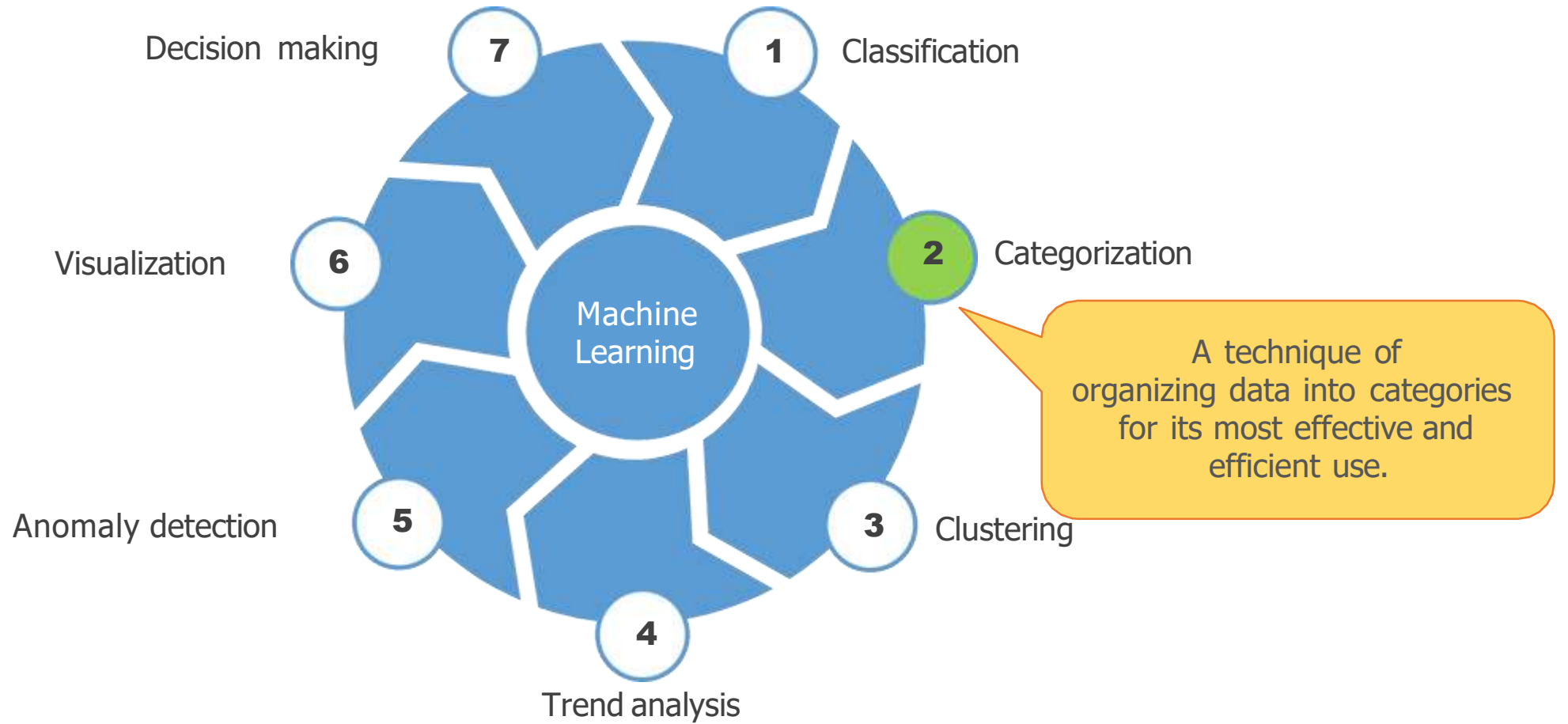


Machine Learning Techniques

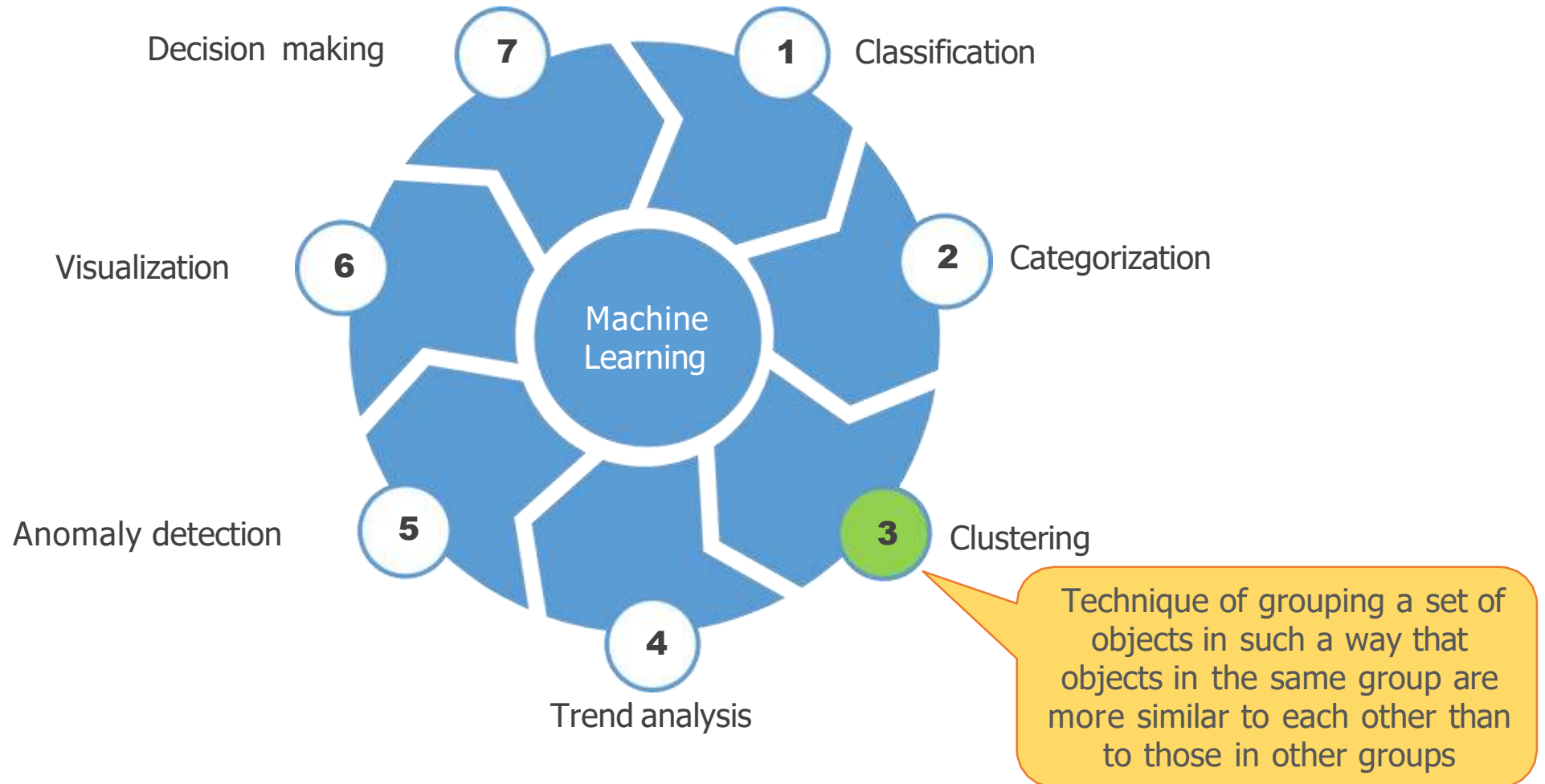


classification is a technique in which the computer program learns from the data input given to it and then uses this **learning** to **classify** new observation

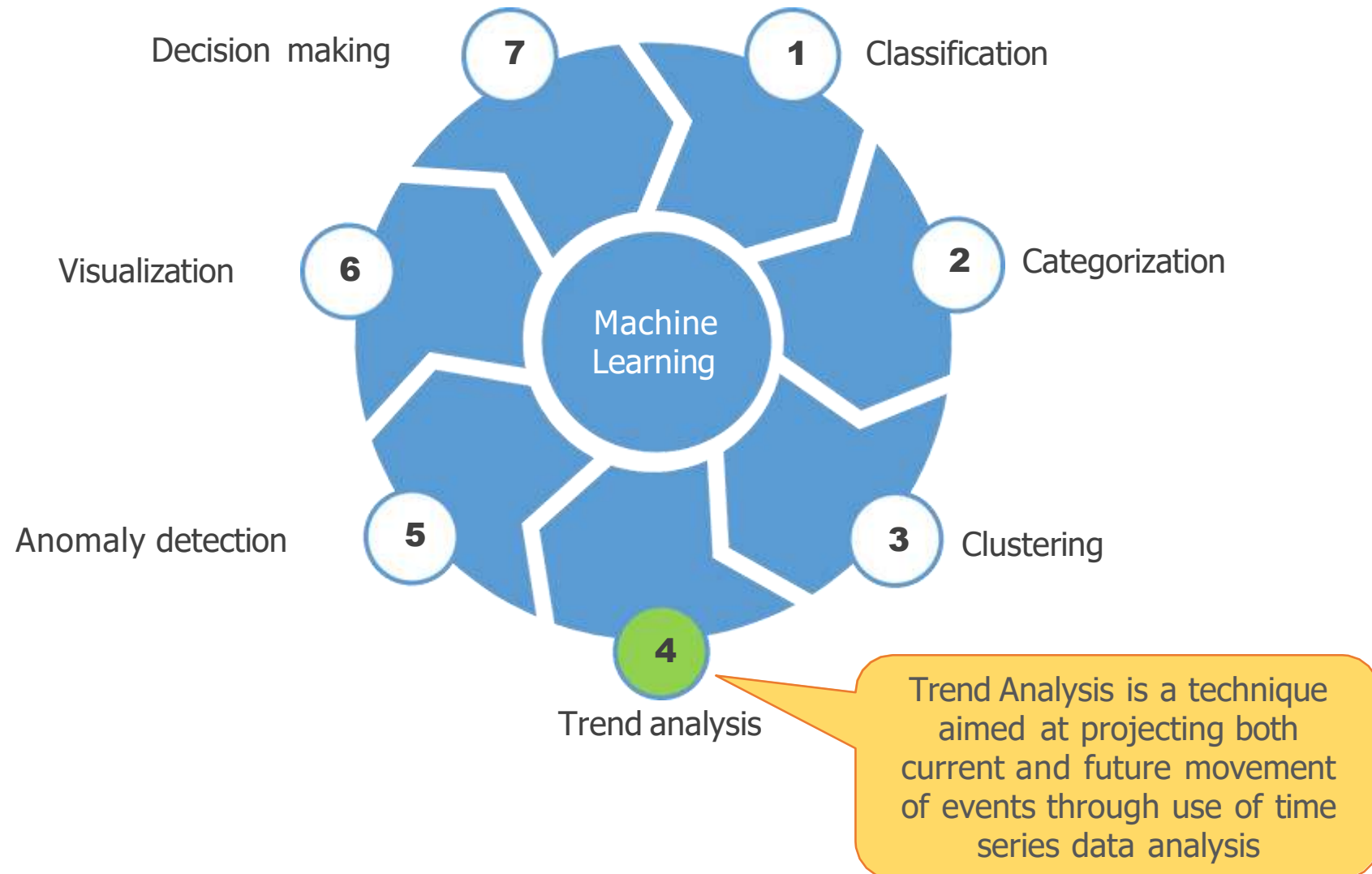
Machine Learning Techniques



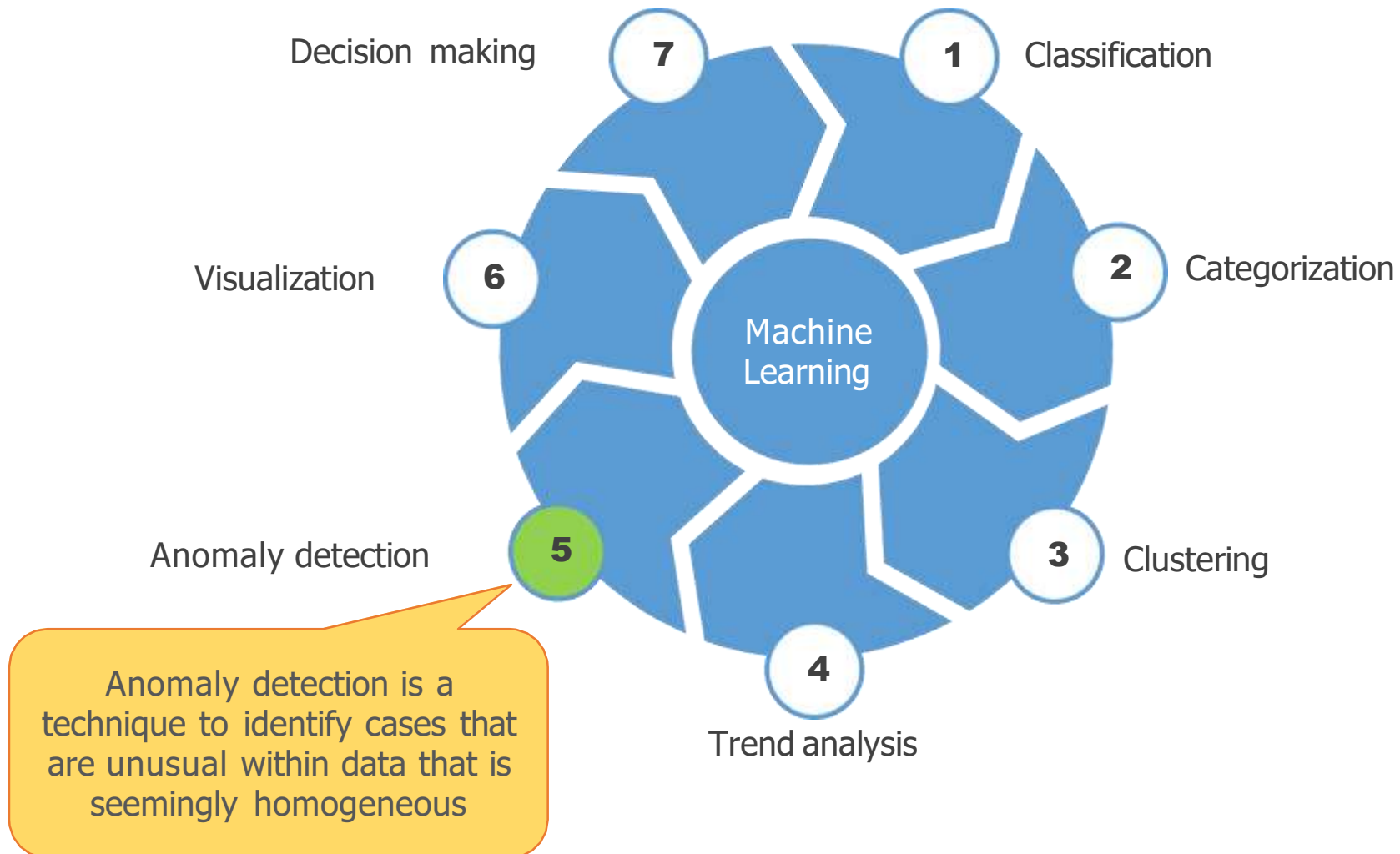
Machine Learning Techniques



Machine Learning Techniques

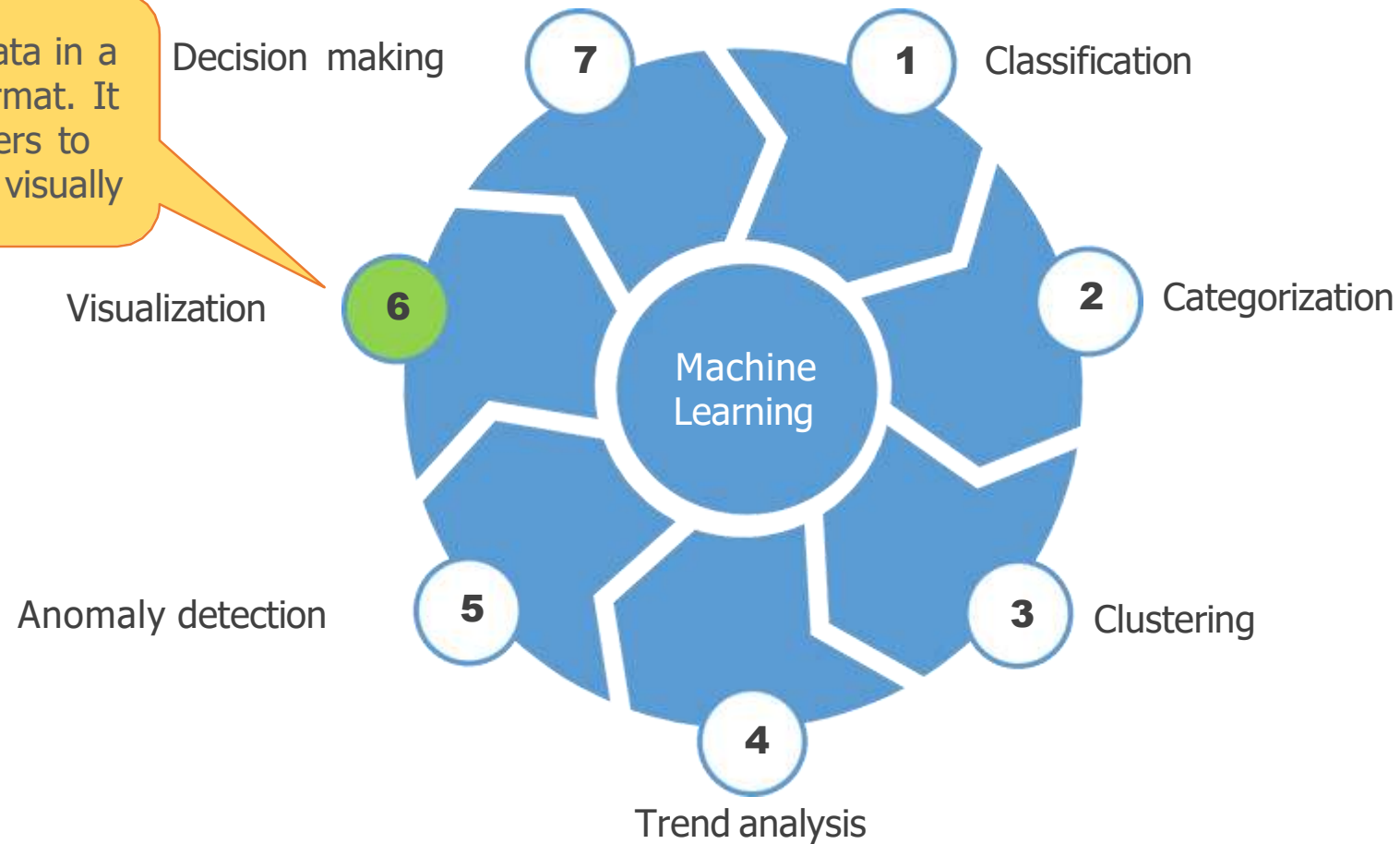


Machine Learning Techniques



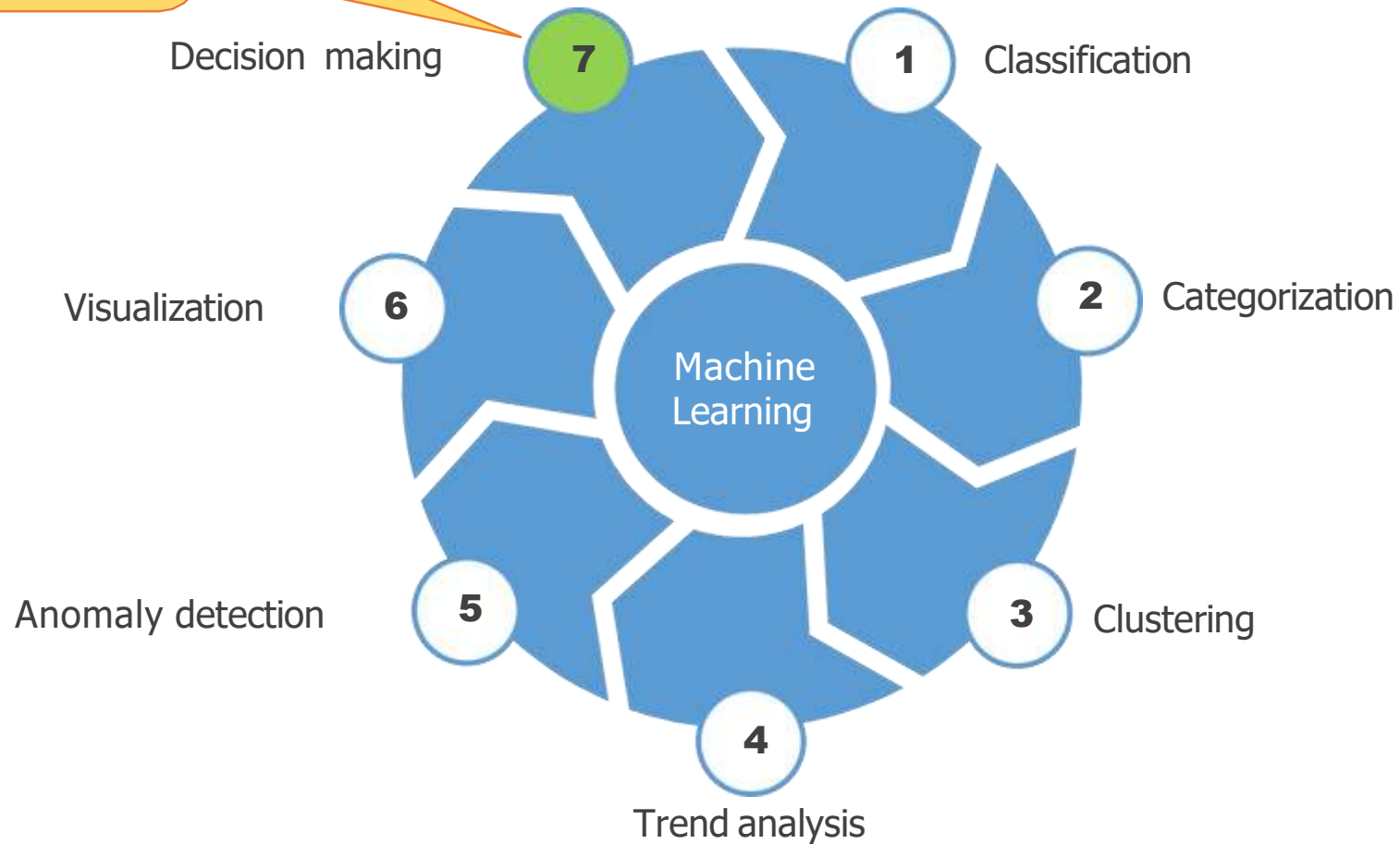
Machine Learning Techniques

Technique to present data in a pictorial or graphical format. It enables decision makers to see analytics presented visually

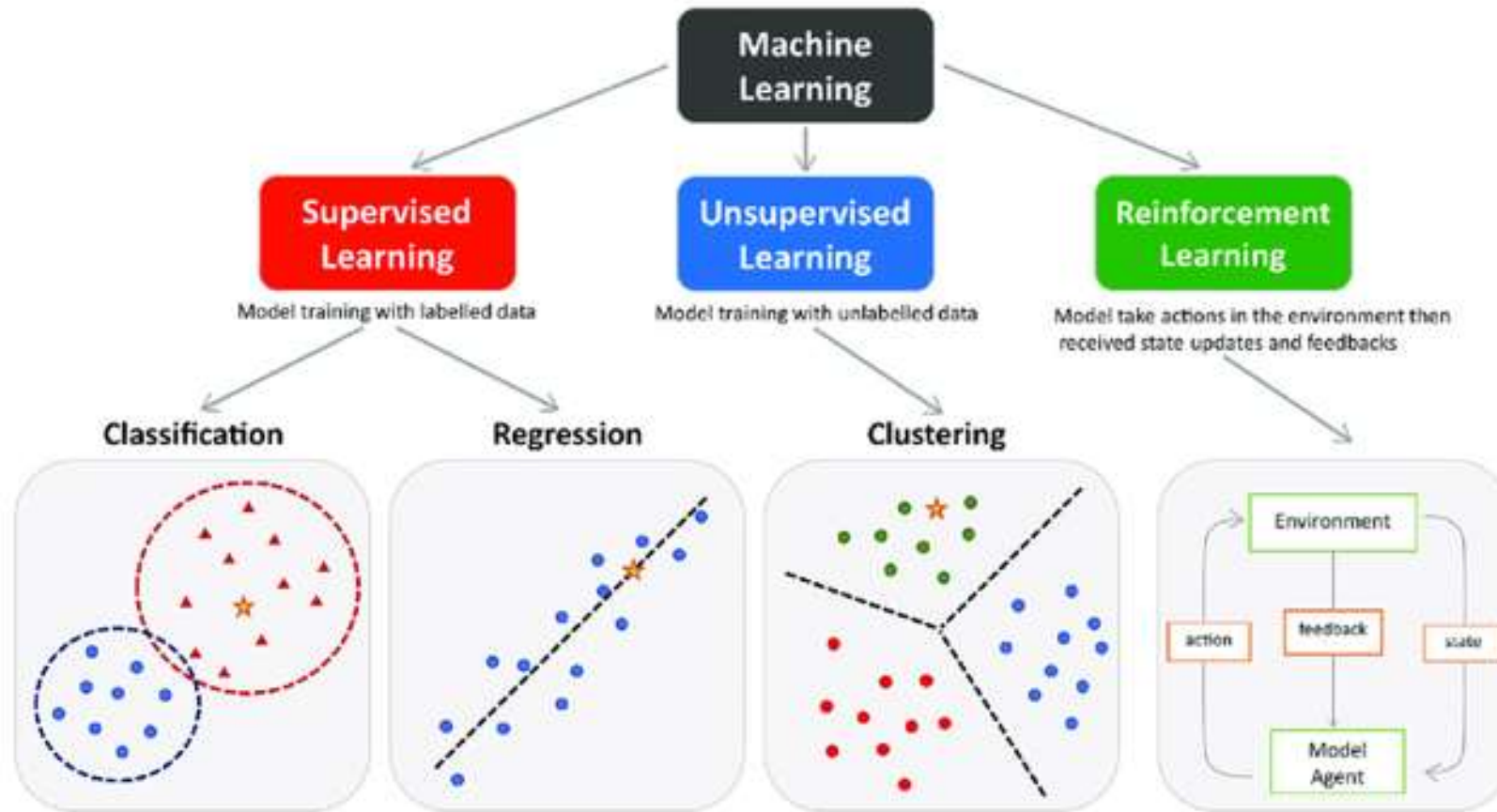


A technique/skill which provides you with the ability to influence managerial decisions with data as evidence for those possibilities

Machine Learning Techniques



Types of Machine Learning



Types of Machine Learning

- **Supervised:** Learning with a **labeled training** set

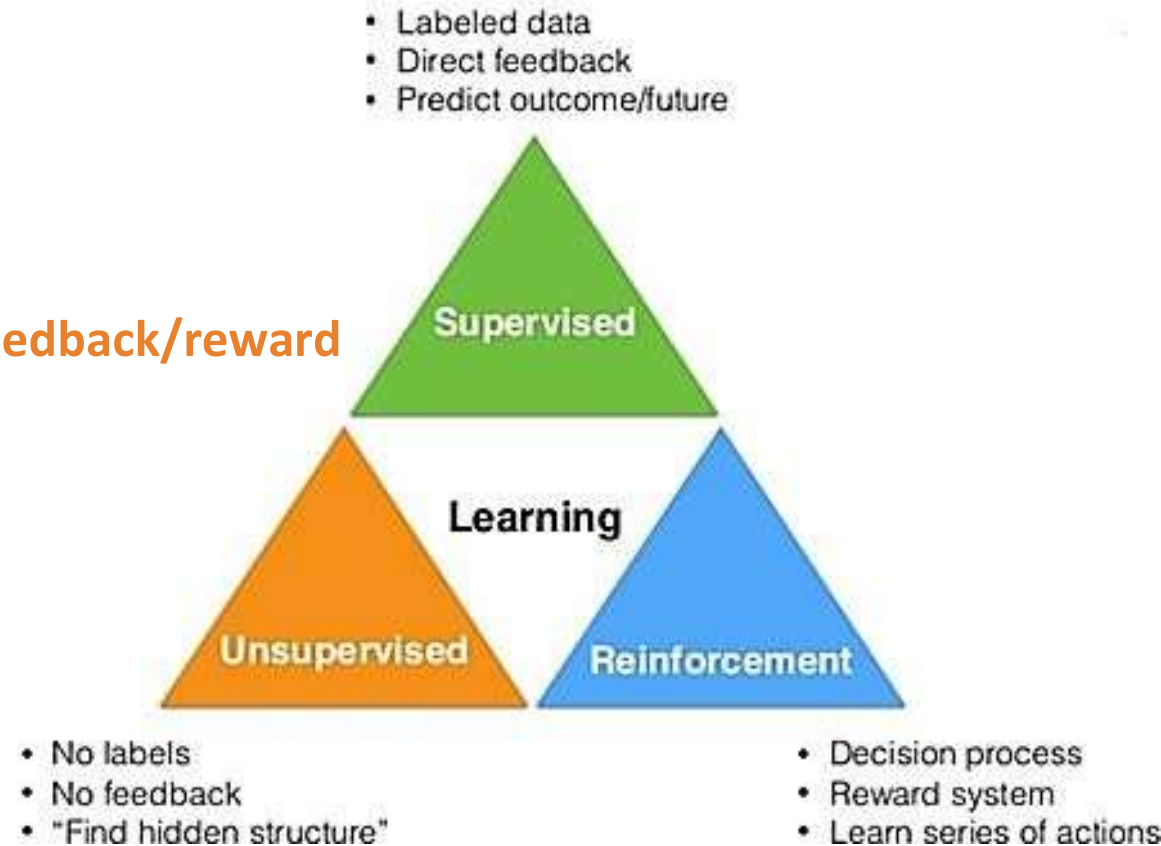
Example: email *classification* with already labeled emails and
Weather forecasting using regression.

- **Unsupervised:** Discover **patterns** in **unlabeled** data

Example: *cluster* similar documents based on text

- **Reinforcement learning:** learn to **act** based on **feedback/reward**

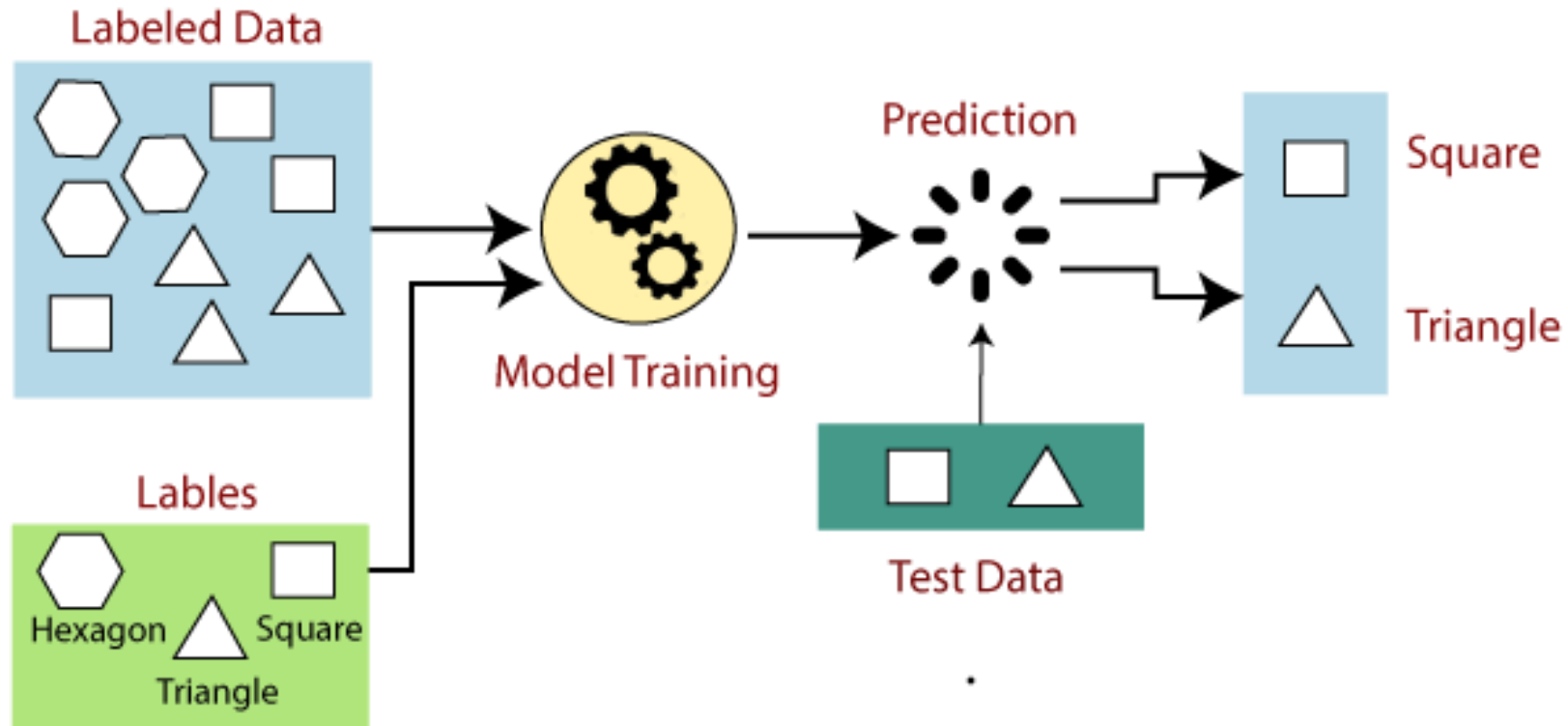
Example: learn to play Go, reward: *win or lose*



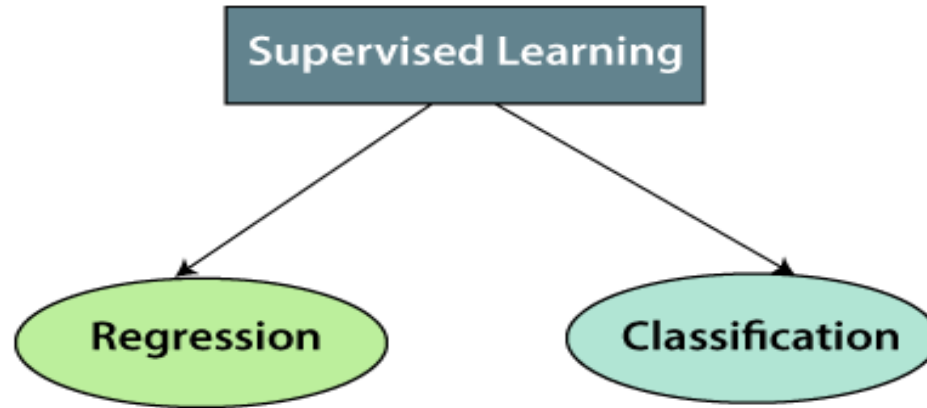
Supervised Learning

- **Supervised learning** is the types of **machine learning** in which machines are trained using well "**labelled**" training data, and on basis of that **data**, machines **predict** the **output**. The labelled data means some input data is already tagged with the correct output.
- In **supervised learning**, the training data provided to the machines work as the **supervisor** that teaches the machines to predict the output correctly.
- The aim of a supervised learning algorithm is to **find a mapping function to map the input variable(x) with the output variable(y)**.

How Supervised Learning Works?



Types of supervised Machine learning Algorithms:



- **Regression** is used to understand the relationship between dependent and independent variables. It is commonly used to make projections, such as for sales revenue for a given business. Linear regression, logistical regression and polynomial regression are popular regression algorithms.
- **Regression** algorithms are used if there is a relationship between the input variable and the output variable. It is used for the prediction of continuous variables.

Types of supervised Machine learning Algorithms:

❑ Classification:

- Inputs are divided into two or more classes, and the learner must produce a model that assigns unseen inputs to one or more (multi-label classification) of these classes.
- Classification algorithms are used when the output variable is categorical, which means there are two classes such as Yes-No, Male-Female, True-false, etc.

❑ Types of Classification Algorithms:

- Random Forest
- Decision Trees
- Naïve Bayes
- Support vector Machines
- K- Nearest Neighbors

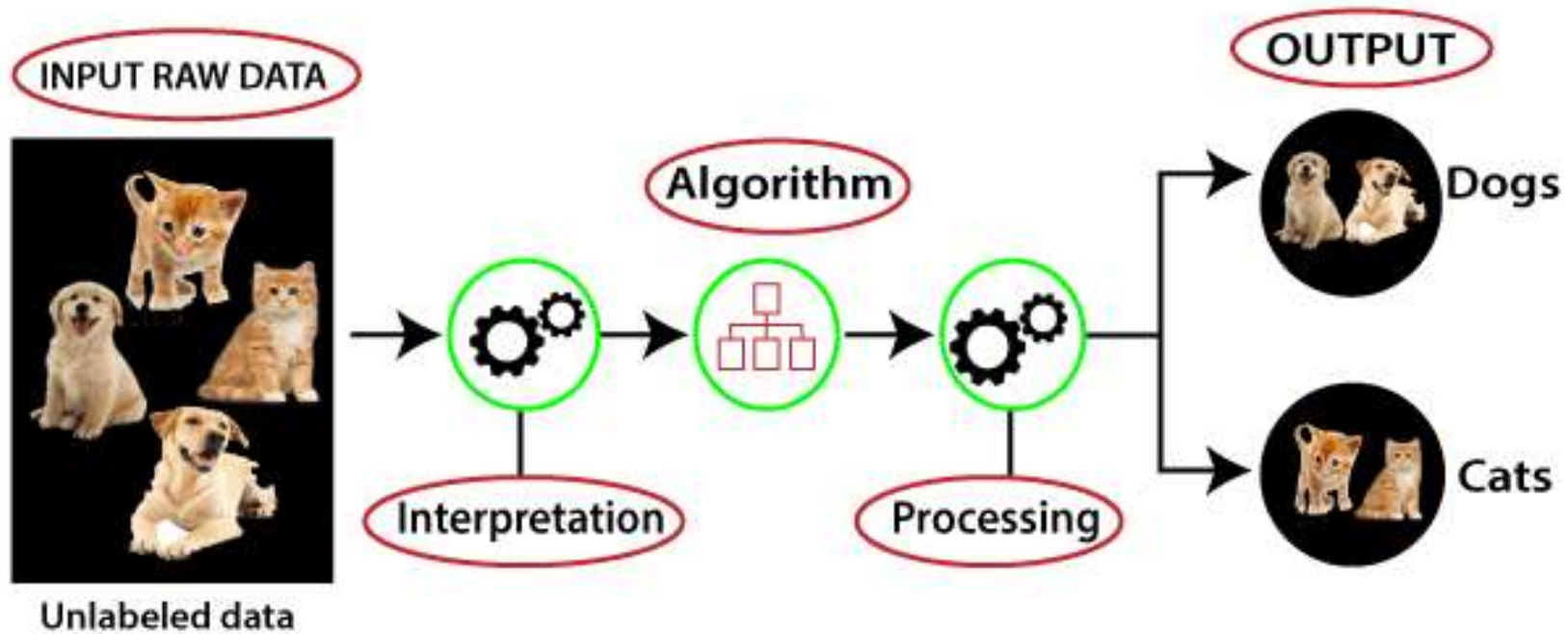
Unsupervised Learning

- **Unsupervised learning** is a type of machine learning in which models are trained using unlabeled dataset and are allowed to act on that data without any supervision.
- **Unsupervised learning** is a machine learning technique in which models are not supervised using training dataset. Instead, models itself find the hidden patterns and insights from the given data.
- The goal of unsupervised learning is to **find the underlying structure of dataset, group that data according to similarities, and represent that dataset in a compressed format**

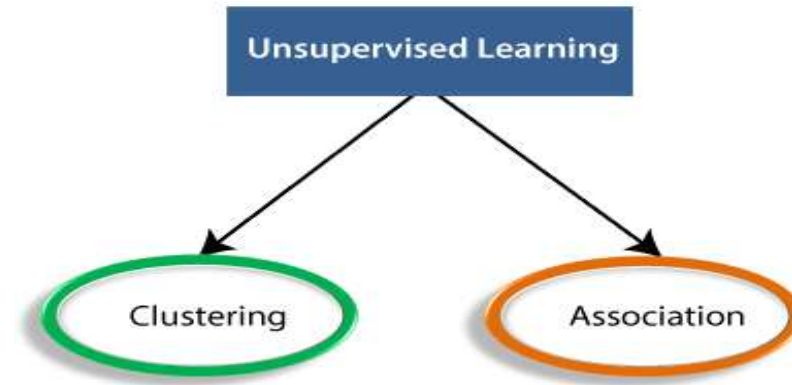
Need of Unsupervised Learning

- Unsupervised learning is helpful for finding useful insights from the data.
- Unsupervised learning is much similar as a human learns to think by their own experiences, which makes it closer to the real AI.
- Unsupervised learning works on unlabeled and uncategorized data which make unsupervised learning more important.

Working of Unsupervised Learning



Types of Unsupervised Learning Algorithm



❑ **Clustering:** Clustering is a method of grouping the objects into clusters such that objects with most similarities remain in a group and have less or no similarities with the objects of another group.

- K-means clustering
- Partitioning Clustering
- Density-Based Clustering
- Distribution Model-Based Clustering
- Hierarchical Clustering
- Fuzzy Clustering
- Principle Component Analysis

Types of Unsupervised Learning Algorithm

❑ **Association:** An association is an unsupervised learning method which is used for finding the relationships between variables in the large database.

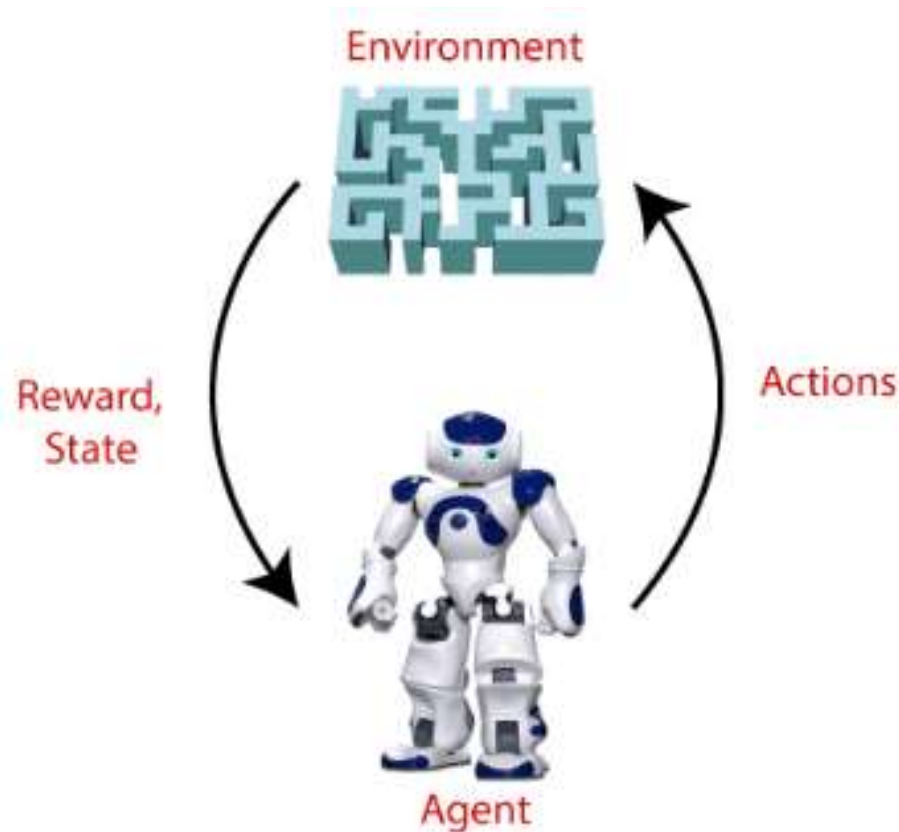
- It determines the set of items that occurs together in the dataset.

❑ Types of Algorithms:

- Apriori Algorithm
- Eclat Algorithm (**Equivalence Class Transformation**)
- F-P Growth Algorithm

Reinforcement Learning

- ❑ Reinforcement Learning(RL) is a type of machine learning technique that enables an agent to learn in an interactive environment by trial and error using feedback from its own actions and experiences.



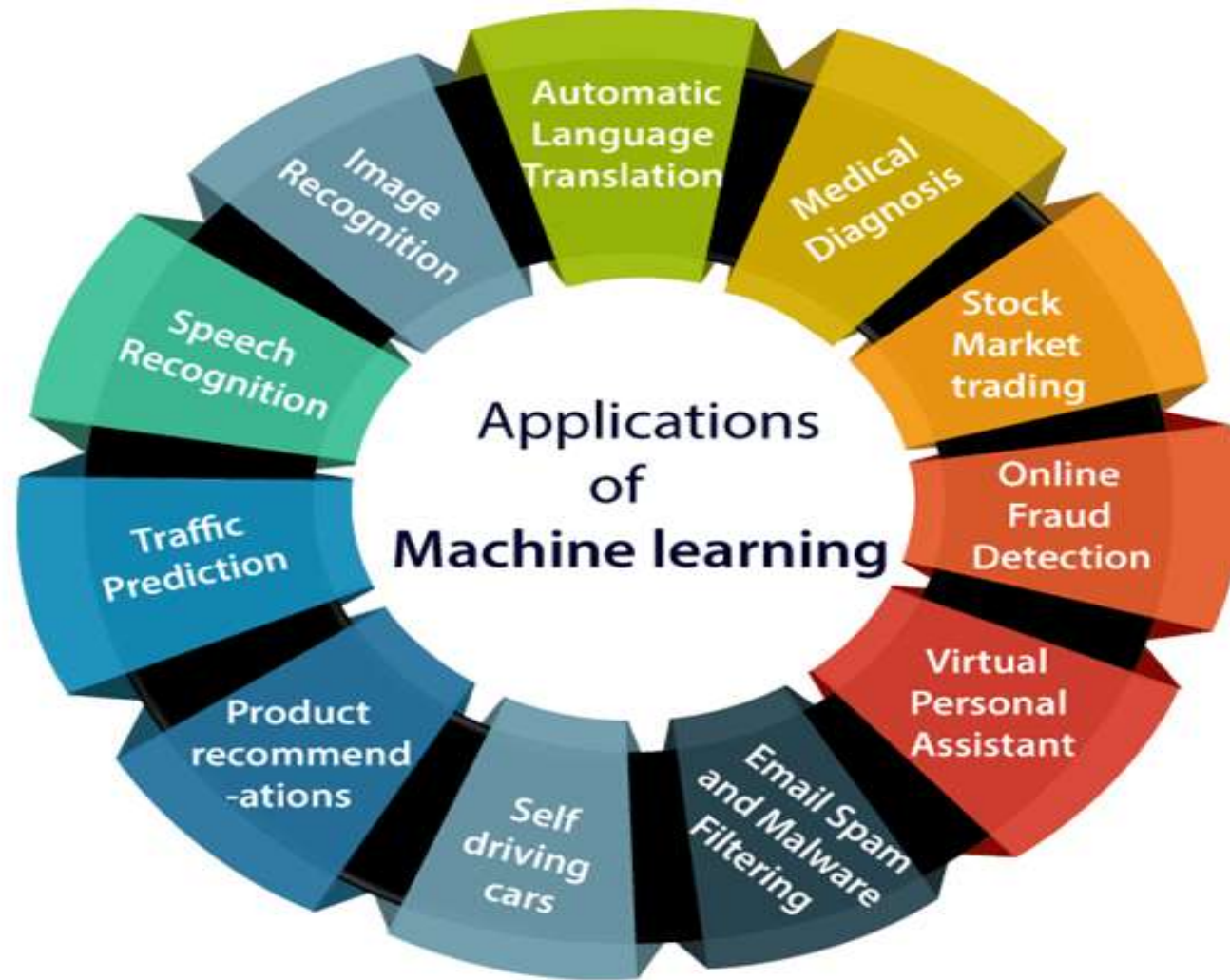
Issues in Machine Learning

- Inadequate Training Data: Noisy data, Incorrect Data.
- Poor quality of data
- Overfitting and Underfitting:
 - ❑ **Overfitting:** It is one of the most common issues faced by Machine Learning engineers and data scientists. Whenever a machine learning model is trained with a huge amount of data, it starts capturing noise and inaccurate data into the training data set. It negatively affects the performance of the model.
 - ❑ **Underfitting:** Whenever a machine learning model is trained with fewer amounts of data, and as a result, it provides incomplete and inaccurate data and destroys the accuracy of the machine learning model.

Issues in Machine Learning

- Monitoring and maintenance: regular monitoring and maintenance become compulsory.
- Lack of skilled resources: The absence of skilled resources in the form of manpower is also an issue.
- Process Complexity of Machine Learning: The machine learning process is very complex, which is also another major issue faced by machine learning engineers and data scientists.
- Change in algorithm with growth in data

Applications of Machine Learning



Applications of Machine Learning

1. **Image recognition** is one of the most common applications of machine learning. It is used to identify objects, persons, places, digital images, etc.

Example: Facebook auto friend tagging. behind this is machine learning's **face detection** and **recognition algorithm**.

2. **Speech recognition** is a process of converting voice instructions into text, and it is also known as **Speech to text** or **Computer speech recognition Search by voice** it comes under speech recognition.

Example: **Google assistant, Siri, Cortana, and Alexa** are using speech recognition technology to follow the voice instructions.

3. **Traffic prediction:** It predicts the traffic conditions such as whether traffic is cleared, slow-moving, or heavily congested with the help of two ways:

Real Time location of the vehicle form Google Map app and sensors

Average time has taken on past days at the same time.

Applications of Machine Learning

4. Product recommendations: Machine learning is widely used by various e-commerce and entertainment companies such as **Amazon**, **Netflix**, etc., for product recommendation to the user.

5. Self-driving cars: Machine learning plays a significant role in self-driving cars. Tesla, the most popular car manufacturing company is working on self-driving car.

6. Email Spam and Malware Filtering: Whenever we receive a new email, it is filtered automatically as important, normal, and spam. We always receive an important mail in our inbox with the important symbol and spam emails in our spam box.

Some machine learning algorithms such as **Multi-Layer Perceptron**, **Decision tree**, and **Naïve Bayes classifier** are used for email spam filtering and malware detection.

Applications of Machine Learning

7. Virtual Personal Assistant: various virtual personal assistants such as **Google assistant, Alexa, Cortana, Siri**. As the name suggests, they help us in finding the information using our voice instruction.

8. Online Fraud Detection: Machine learning is making our online transaction safe and secure by detecting fraud transaction. **Feed Forward Neural network** helps us by checking whether it is a genuine transaction or a fraud.

9. Stock Market trading: In the stock market, there is always a risk of up and downs in shares, so for this machine learning's **long short term memory neural network** is used for the prediction of stock market trends.

Applications of Machine Learning

10. Medical Diagnosis: In medical science, machine learning is used for diseases diagnoses. With this, medical technology is growing very fast and able to build 3D models that can predict the exact things.

11. Automatic Language Translation: machine learning helps us by converting the text into our known languages. Google's GNMT (Google Neural Machine Translation) provide this feature, which is a Neural Machine Learning that translates the text into our familiar language, and it called as automatic translation.

Applications of Machine Learning

Domain	Application
Healthcare	Disease Prediction
Banking	Loan Approval
Finance	Fraud Detection
Agriculture	Crop Disease Detection
Retail	Product Recommendation
Manufacturing	Predictive Maintenance
Cyber Security	Intrusion Detection
Education	Student Performance Prediction
Transportation	Self-driving Cars
Entertainment	Netflix & Spotify Recommendation

Steps involved in developing a Machine Learning Application

1. Problem framing
2. Collecting Data
3. Preparing the Data
4. Choosing a Model
5. Training the Model
6. Evaluating the Model
7. Parameter Tuning
8. Making Predictions

1. **Problem framing:** This first step is to frame a machine learning problem in terms of what to predict and what kind of observation data, have to make for those predictions.
2. **Collecting Data:** Next step is to collect the data from a historical database or from open datasets or from any other data sources.
3. **Preparing the Data:** Cleaning the data to remove unwanted data, missing values, rows, and columns, duplicate values, data type conversion, etc. and Visualize the data
4. **Choosing a Model:** A machine learning model determines the output you get after running a machine learning algorithm on the collected data. It is important to choose a model which is relevant to the task at hand.

5. Training the Model: Training is the most important step in machine learning. In training, you pass the prepared data to your machine learning model to find patterns and make predictions.

6. Evaluating the Model: After training your model, you have to check to see how it's performing. This is done by testing the performance of the model on previously unseen data.

7. Parameter Tuning: Parameters are the variables in the model that the programmer generally decides. At a particular value of your parameter, the accuracy will be the maximum. Parameter tuning refers to finding these values.

8. Making Predictions: In the end, you can use your model on unseen data to make predictions accurately.

Hypothesis

A hypothesis is a mathematical function that predicts the output from the input.

Example:

For Linear Regression

$$h(x) = wx + b$$

where:

- $h(x)$: Predicted output
- w : Weight
- b : Bias

Example

Predicting house price from area:

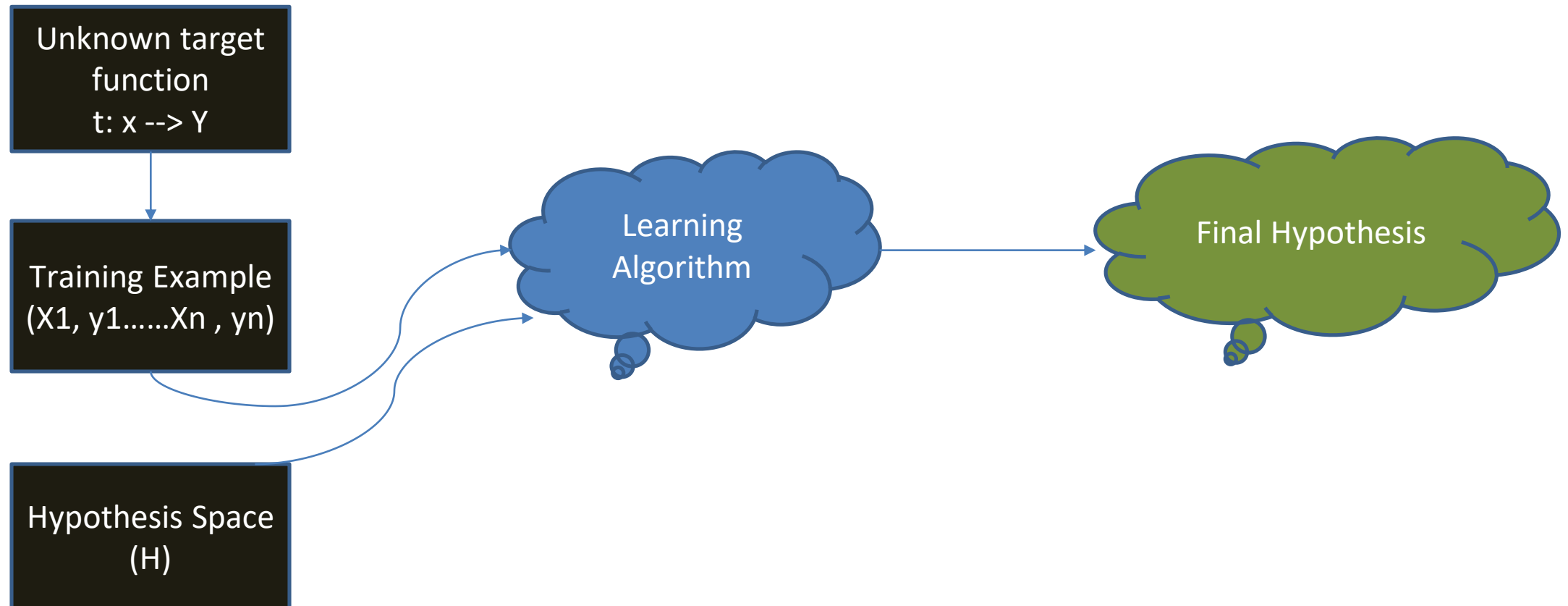
$$\text{Price} = 5000 \times \text{Area} + 200000$$

Hypothesis

- The hypothesis is defined as the **supposition or proposed explanation based on insufficient evidence or assumptions.**
- It is just a guess based on some known facts but has not yet been proven. A good hypothesis is testable, which results in either true or false.
- **Example:** Let's understand the hypothesis with a common example.
 - **Some scientist claims that ultraviolet (UV) light can damage the eyes then it may also cause blindness.**
 - In this example, a scientist just claims that UV rays are harmful to the eyes, but we assume they may cause blindness.
 - However, *it may or may not be possible*. Hence, these types of assumptions are called a hypothesis.

Hypothesis in Machine Learning

- The hypothesis is one of the commonly used concepts of statistics in Machine Learning.
- It is specifically used in Supervised Machine learning, where an ML model learns a function that best maps the input to corresponding outputs with the help of an available dataset.



Hypothesis in Machine Learning Cont....

- **Hypothesis Space (H):**
- Hypothesis space is defined as a **set of all possible legal hypotheses**; hence it is also known as a hypothesis set. It is used by supervised machine learning algorithms to determine the best possible hypothesis to describe the target function or best maps input to output.

Hypothesis in Machine Learning Cont....

- **Hypothesis (h):**

- It is defined as the *approximate function* that best describes the target in supervised machine learning algorithms. It is primarily based on data as well as bias and restrictions applied to data.
- Hence A hypothesis (h) can be concluded as a single hypothesis that maps input to proper output. It can be evaluated as well as used to make predictions.
- The hypothesis (h) can be formulated in machine learning as follows:

$$y = mx + c$$

Where

y = Predicted output

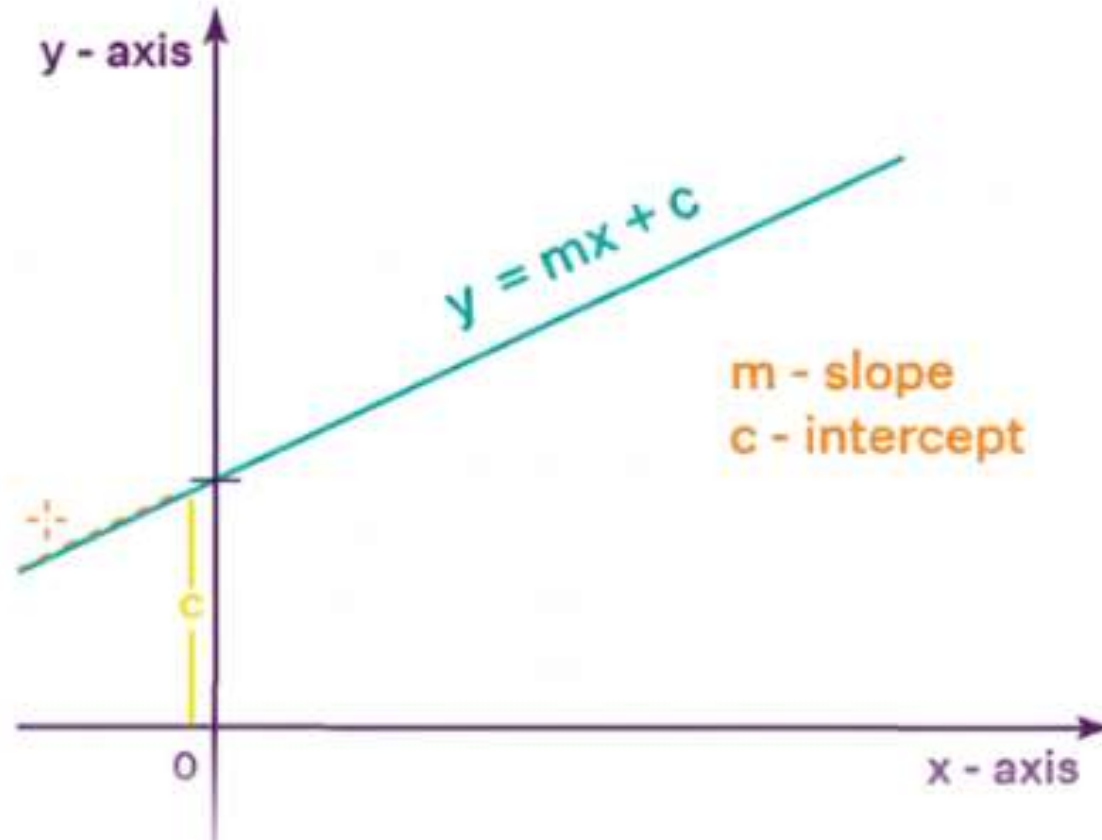
m= Slope of the line which divided test data or changes in y divided by change in x

x= Input

c= Intercept (Constant)

Hypothesis in Machine Learning Cont....

- Slope intercept Form: $y=mx+c$



Types of hypotheses

- **Null Hypothesis:**

- Null Hypothesis states that there is no significant difference/ relation between the two samples.
- It is usually denoted by symbol H_0 .
- The difference in the sample is just due to fluctuation of sampling
- There is no difference between mean of two groups.
- **Example:** If a scientist is studying the effect of a new drug on blood pressure, the null hypothesis might state that the **drug has no impact on blood pressure.**
- Imagine a study to determine whether a new teaching method improves math scores, The null hypothesis would state, “ The new teaching method has no effect on math scores compared to the traditional method.”

Types of hypotheses

- **Alternative Hypothesis:**

- Alternative Hypothesis predicts there is relationship **between the two variables.**
- It is usually denoted by symbol H_1 or H_a .
- **Example:** Using the same example of a drug study, the alternative hypothesis might state that the **drug does reduce blood pressure**

Inductive Bias

- Inductive bias is the set of assumptions a learning algorithm makes to generalize from training data to unseen data.
- These biases can influence the model's ability to learn from a given dataset and can affect the performance of the model on new, unseen data.
- It is important to consider the inductive bias of a model when selecting or designing a machine learning algorithm, because it can have a significant impact on the model's performance.
- For example, a linear regression model has an inductive bias towards linear relationships between variables, while a decision tree has an inductive bias towards creating simple, hierarchical partitions of the data.
- "The inductive bias (also known as learning bias) of a learning algorithm is the set of assumptions that the learner uses to predict outputs of given inputs".

Bias-Variance Tradeoff

- The **bias-variance tradeoff** is a fundamental concept in machine learning that helps us to understand the performance of a model.
- It describes the relationship between a model's complexity, its ability to generalize to new data, and the errors it makes.
- **Bias:**
 - **Definition:** Error due to overly simplistic assumptions in the learning algorithm.
 - **Result:** The model **underfits** the data, failing to capture underlying patterns.
 - **Example:** Using a linear model to fit non-linear data.
- **Variance:**
 - **Definition:** Error due to the model's sensitivity to small fluctuations in the training set.
 - **Result:** The model **overfits** the data, capturing noise instead of the underlying pattern.
 - **Example:** Using a high-degree polynomial to fit a small dataset.
- **Tradeoff:**
 - Increasing model complexity reduces bias but increases variance.
 - Decreasing model complexity reduces variance but increases bias.
 - The goal is to find the **optimal balance** where both bias and variance are minimized.

Bias-Variance Tradeoff

❑ Visual Representation

➤ High Bias (Underfitting):

- Training error: High
- Validation error: High
- Model is too simple to capture the data's complexity.

➤ High Variance (Overfitting):

- Training error: Low
- Validation error: High
- Model is too complex and fits noise in the training data.

➤ Optimal Model:

- Training error: Low
- Validation error: Low
- Model generalizes well to unseen data.

Model Complexity	Bias	Variance	Performance
Very Simple	High	Low	Underfitting
Moderate	Low	Low	Best Model
Very Complex	Low	High	Overfitting

Bias-Variance Tradeoff

➤ Solutions for High Bias (Underfitting)

❑ Increase Model Complexity:

- Use a more complex model (e.g., switch from linear to polynomial regression).
- Add more features or interactions between features.

❑ Reduce Regularization:

- Decrease regularization strength (e.g., reduce λ in L2 regularization).

❑ Train Longer:

- For iterative algorithms like gradient descent, train for more epochs.

Model Complexity	Bias	Variance	Performance
Very Simple	High	Low	Underfitting
Moderate	Low	Low	Best Model
Very Complex	Low	High	Overfitting

Bias-Variance Tradeoff

➤ **Solutions for High Variance (Overfitting)**

❑ **Get More Data:**

- Collect more training data to reduce overfitting.

❑ **Simplify the Model:**

- Use a simpler model (e.g., reduce the degree of a polynomial).
- Reduce the number of features (feature selection).

❑ **Regularization:**

- Add regularization terms (e.g., L1/L2 regularization) to penalize complexity.

❑ **Cross-Validation:**

- Use k-fold cross-validation to ensure the model generalizes well.

❑ **Early Stopping:**

- Stop training when validation error starts increasing.

Bias-Variance Tradeoff

❑ Practical Example:

➤ Scenario:

- You are building a model to predict house prices.
- Training error is low, but validation error is high.

➤ Diagnosis:

- High variance (overfitting).

❑ Solutions:

➤ Regularization:

- Add L2 regularization to the loss function.

➤ Feature Selection:

- Remove irrelevant features (e.g., "number of windows").

➤ More Data:

- Collect more house price data.

➤ Simplify Model:

- Use a simpler model (e.g., linear regression instead of a neural network).

Performance measures

□ Performance measures evaluate the effectiveness of Machine Learning models.

1. Regression Metrics

I. **Mean Absolute Error (MAE):** Average absolute difference between actual and predicted values.

$$MAE = \frac{1}{n} \sum |y - \hat{y}|$$

II. **Mean Squared Error (MSE):** Average of squared prediction errors.

$$MSE = \frac{1}{n} \sum (y - \hat{y})^2$$

III. **Root Mean Squared Error (RMSE):** Square root of MSE.

$$RMSE = \sqrt{MSE}$$

IV. **R² Score:** Measures how well the model explains the variation in the data. It will ranges between 0 to 1 . Higher values indicate a better fit.

Performance measures continue..

❑ Performance measures evaluate the effectiveness of Machine Learning models.

2. Classification Metrics

➤ Confusion Matrix

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

1. **Accuracy:** Accuracy measures the **overall correctness** of the model. It tells us: Out of all predictions, how many predictions are correct?

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Example: Suppose TP = 80 , TN = 10 , FP = 5 , FN = 5

Accuracy = $\frac{80+10}{80+10+5+5} = 90\%$

Interpretation

The model predicts **90% of samples correctly**.

2. Precision: Precision is the ratio between the True Positives and all the Positives. It shows how many of the “**yes**” predictions made by the model were actually correct. It helps us reduce wrong “yes” guesses which are called false positives (FP). **Precision tells us: Out of all samples predicted as Positive, how many are actually Positive?**

$$Precision = \frac{TP}{TP + FP}$$

Example: Suppose TP = 90 , FP = 10 Then Precision=90/90+10=90%

Interpretation

Out of every **100 predicted positive cases, 90 are actually positive.**

Applications

Precision is important when **False Positives are costly.**

Examples:

- Spam Email Detection
- Fraud Detection

3. Recall (Sensitivity): Measures how good the model is at predicting positives. It shows the proportion of true positives detected out of all the actual positive instances. High recall is essential when missing positive cases has significant consequences like in medical tests. **Recall tells us: Out of all actual Positive samples, how many did the model correctly identify?**

$$Recall = \frac{TP}{TP + FN}$$

Example: Suppose $TP = 90$, $FN = 10$

Then

$Recall = 90 / (90 + 10) = 90\%$

Interpretation

The model correctly detects **90% of actual positive cases**.

Applications

Recall is important when **False Negatives are dangerous**.

Examples:

- Cancer Detection
- COVID Diagnosis
- Fraud Detection

4. F1-Score :It combines precision and recall into a single metric to balance their trade-off. It provides a better sense of a model's overall performance particularly for imbalanced datasets. It is helpful when both false positives and false negatives are important though it assumes precision and recall are equally important but in some situations one might matter more than the other.

- F1-Score is the harmonic mean of Precision and Recall.
- It provides a single measure that balances both.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Example Suppose

- Precision = 90%
- Recall = 80%

Then

$$F1 = 2 \times 0.90 \times 0.80 / 0.90 + 0.8 = \mathbf{0.847}$$

Interpretation

Higher F1-Score means the model has a good balance between Precision and Recall.

5. Specificity: Specificity tells us: Out of all actual Negative samples, how many were correctly predicted as Negative?

$$\text{Specificity} = \frac{TN}{TN + FP}$$

Example: Suppose

- TN = 80

- FP = 20

Then

Specificity=80/80+20=

Interpretation

The model correctly identifies **80% of negative cases**.

Applications

Specificity is important when avoiding **False Positives**.

Example: A healthy patient should not be incorrectly diagnosed as having a disease.

Metric	Formula	What It Measures	Best Used When
Accuracy	$(\frac{TP+TN}{TP+TN+FP+FN})$	Overall correctness	Classes are balanced
Precision	$(\frac{TP}{TP+FP})$	Correctness of positive predictions	False Positives are costly (Spam, Fraud)
Recall (Sensitivity)	$(\frac{TP}{TP+FN})$	Ability to detect actual positives	False Negatives are costly (Cancer Detection)
Specificity	$(\frac{TN}{TN+FP})$	Ability to detect actual negatives	Avoiding False Positives
F1-Score	$(\frac{2PR}{P+R})$	Balance between Precision and Recall	Imbalanced datasets

Metric	Easy Question to Remember
Accuracy	How many predictions are correct overall?
Precision	Of all predicted YES, how many are actually YES?
Recall	Of all actual YES, how many did the model find?
Specificity	Of all actual NO, how many did the model correctly identify?
F1-Score	How well are Precision and Recall balanced?

Problem-I: Calculate the Accuracy, Precision, Recall and F1 score for the following confusion matrix.

	Predicted No	Predicted Yes
Actual No	50	10
Actual Yes	5	100

Given

	Predicted No	Predicted Yes
Actual No	50	10
Actual Yes	5	100

Step 1: Identify TP, TN, FP, FN

- True Positive (TP) = 100
- True Negative (TN) = 50
- False Positive (FP) = 10
- False Negative (FN) = 5

Total samples:
TP+TN+FP+FN=100+50+10+5=**165**

Metric	Formula	Value
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	90.91%
Precision	$\frac{TP}{TP+FP}$	90.91%
Recall	$\frac{TP}{TP+FN}$	95.24%
F1-Score	$\frac{2PR}{P+R}$	93.02%

Choice of a performance measure

Performance Measure	Used For	Learning Type	Output Type	Description
MAE (Mean Absolute Error)	Regression	Supervised Learning	Continuous values	Measures the average absolute difference between actual and predicted values. Lower is better.
MSE (Mean Squared Error)	Regression	Supervised Learning	Continuous values	Measures the average squared error. Penalizes large errors more heavily. Lower is better.
RMSE (Root Mean Squared Error)	Regression	Supervised Learning	Continuous values	Square root of MSE. Error is expressed in the same units as the target variable. Lower is better.
R² Score (Coefficient of Determination)	Regression	Supervised Learning	Continuous values	Measures how well the model explains the variation in the data. Higher is better (maximum = 1).
Accuracy	Classification	Supervised Learning	Categorical/Class labels	Percentage of correctly classified instances. Best for balanced datasets.
Precision	Classification	Supervised Learning	Categorical/Class labels	Measures how many predicted positive instances are actually positive. Important when false positives are costly.
Recall (Sensitivity)	Classification	Supervised Learning	Categorical/Class labels	Measures how many actual positive instances are correctly identified. Important when false negatives are costly.
F1-Score	Classification	Supervised Learning	Categorical/Class labels	Harmonic mean of Precision and Recall. Useful for imbalanced datasets.

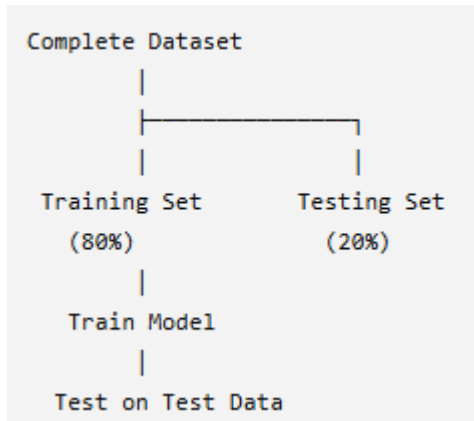
Algorithm	Task	Performance Measures
Linear Regression	Regression	MAE, MSE, RMSE, R ²
Polynomial Regression	Regression	MAE, MSE, RMSE, R ²
Ridge Regression	Regression	MAE, MSE, RMSE, R ²
Lasso Regression	Regression	MAE, MSE, RMSE, R ²
Logistic Regression	Classification	Accuracy, Precision, Recall, F1-Score, ROC-AUC
Decision Tree (Classifier)	Classification	Accuracy, Precision, Recall, F1-Score
Random Forest (Classifier)	Classification	Accuracy, Precision, Recall, F1-Score
Support Vector Machine (Classifier)	Classification	Accuracy, Precision, Recall, F1-Score
Naïve Bayes	Classification	Accuracy, Precision, Recall, F1-Score
K-Nearest Neighbors (Classifier)	Classification	Accuracy, Precision, Recall, F1-Score

Data Validation Techniques in Machine Learning

- ❑ Data validation techniques are used to **evaluate the performance of a Machine Learning model** and ensure that it performs well on **unseen data**. They help to detect **overfitting** and **underfitting**.

1. Holdout Method: The **Holdout Method** is the simplest validation technique where the dataset is divided into **two parts**:

- **Training Set** (70–80%)
- **Testing Set** (20–30%)



Advantages

- Simple and easy to implement.
- Fast execution.
- Suitable for large datasets.

Disadvantages

- Performance depends on a single random split.
- Some data may never be used for training.
- Results may vary with different splits.

Applications

- Large datasets
- Initial model evaluation

2. Random Subsampling (Repeated Holdout)

Random Subsampling is an extension of the Holdout Method. Instead of splitting the data only once, the dataset is randomly split **multiple times**, and the model performance is averaged.

Dataset

Split 1 → Train → Test → Accuracy1

Split 2 → Train → Test → Accuracy2

Split 3 → Train → Test → Accuracy3

Average Accuracy

Advantages

- More reliable than a single Holdout Method.
- Reduces the effect of random data splitting.

Disadvantages

- Some samples may never be selected for testing.
- Computationally more expensive than Holdout.

Applications

- Medium-sized datasets
- Comparing different models

3. Cross Validation

Cross Validation evaluates the model by dividing the dataset into **K equal folds**. Each fold is used once as the testing set while the remaining folds are used for training.

```
5-Fold Cross Validation

Fold 1 → Test
Fold 2 → Train
Fold 3 → Train
Fold 4 → Train
Fold 5 → Train

Repeat until every fold becomes the test set.
```

Types

- K-Fold Cross Validation
- Stratified K-Fold
- Leave-One-Out Cross Validation (LOOCV)

Advantages

- Uses all data for training and testing.
- More reliable than Holdout.
- Reduces overfitting.
- Provides stable performance estimates.

Disadvantages

- Computationally expensive.
- Slower for large datasets.

Applications

- Model evaluation
- Hyper parameter tuning
- Comparing algorithms

4. Bootstrap

Bootstrap is a **resampling technique** in which samples are selected **with replacement**. **With replacement** means that after a sample is selected, it is placed back into the dataset and can be selected again.

Original Dataset

A B C D E

Bootstrap Sample

A C C E B

Notice that:

- C appears twice.
- D is not selected.

Advantages

- Useful for small datasets.
- Estimates model stability.
- Reduces variance.
- Basis of ensemble methods like Random Forest (Bagging).

Disadvantages

- Some records are repeated.
- Some records may never be selected.
- Computationally intensive.

Applications

- Small datasets
- Confidence interval estimation
- Random Forest
- Ensemble Learning

Summary

Technique	Main Purpose	Best Use
Holdout Method	Simple train-test split	Quick evaluation on large datasets
Random Subsampling	Repeated holdout with averaging	More reliable evaluation than a single holdout
Cross Validation	Uses all data for training and testing	Model evaluation, model selection, hyperparameter tuning
Bootstrap	Resampling with replacement	Small datasets, confidence estimation, Random Forest

Model Selection Using Statistical Tests of Significance

When multiple models achieve similar performance, statistical tests determine whether the observed differences are significant.

❑ Common Statistical Tests

- Paired t-test
- McNamara's Test
- Wilcoxon Signed-Rank Test
- ANOVA

❑ Purpose

- Compare competing models objectively.
- Select the model with statistically superior performance rather than relying only on accuracy.

Comparing Classifiers Based on Cost–Benefit

Different types of classification errors may have different costs.

Example: Disease Diagnosis

- ❑ **False Positive (FP):** A healthy person is incorrectly diagnosed as having the disease.
- ❑ **False Negative (FN):** A patient with the disease is incorrectly classified as healthy.

In medical diagnosis, a **False Negative** is usually much more serious because it may delay treatment. Therefore, the preferred classifier is one that minimizes costly errors, even if its overall accuracy is slightly lower.

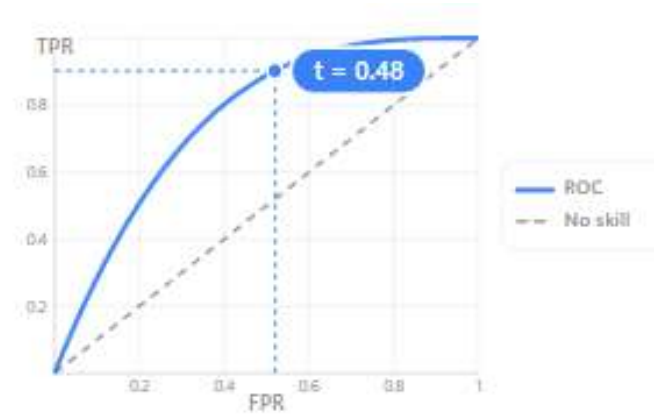
ROC Curve (Receiver Operating Characteristic Curve)

- ❑ The **ROC Curve** is one of the most important performance evaluation metrics for **binary classification** models.
- ❑ It helps us understand **how well a classifier distinguishes between two classes (Positive and Negative)** at different classification thresholds.
- ❑ It was originally developed during World War II for radar signal detection and is now widely used in Machine Learning and medical diagnosis.
- ❑ The ROC Curve is a graph that shows the relationship between: True Positive Rate (TPR) → Recall (Sensitivity) & False Positive Rate (FPR) at different threshold values.

Why Do We Need ROC Curve?

Suppose a Logistic Regression model predicts probabilities.

Patient	Probability of Disease
A	0.95
B	0.82
C	0.70
D	0.45
E	0.20



If we use a threshold of **0.5**:

$\geq 0.5 \rightarrow$ Disease

$< 0.5 \rightarrow$ Healthy

But what if we change the threshold to **0.3** or **0.7**?

The number of True Positives and False Positives changes.

The ROC Curve helps us evaluate model performance **for all possible thresholds**, not just one.

At threshold 0.48, the true-positive rate is 90.2% and the false-positive rate is 52%; AUROC 0.76 indicates useful separation

AUROC 0.76

Threshold 0.48

Thank You..